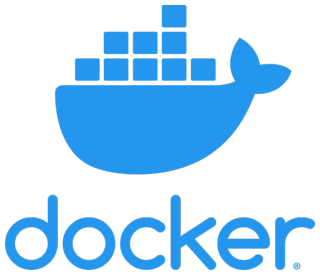
Containers on AWS

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What is Docker?

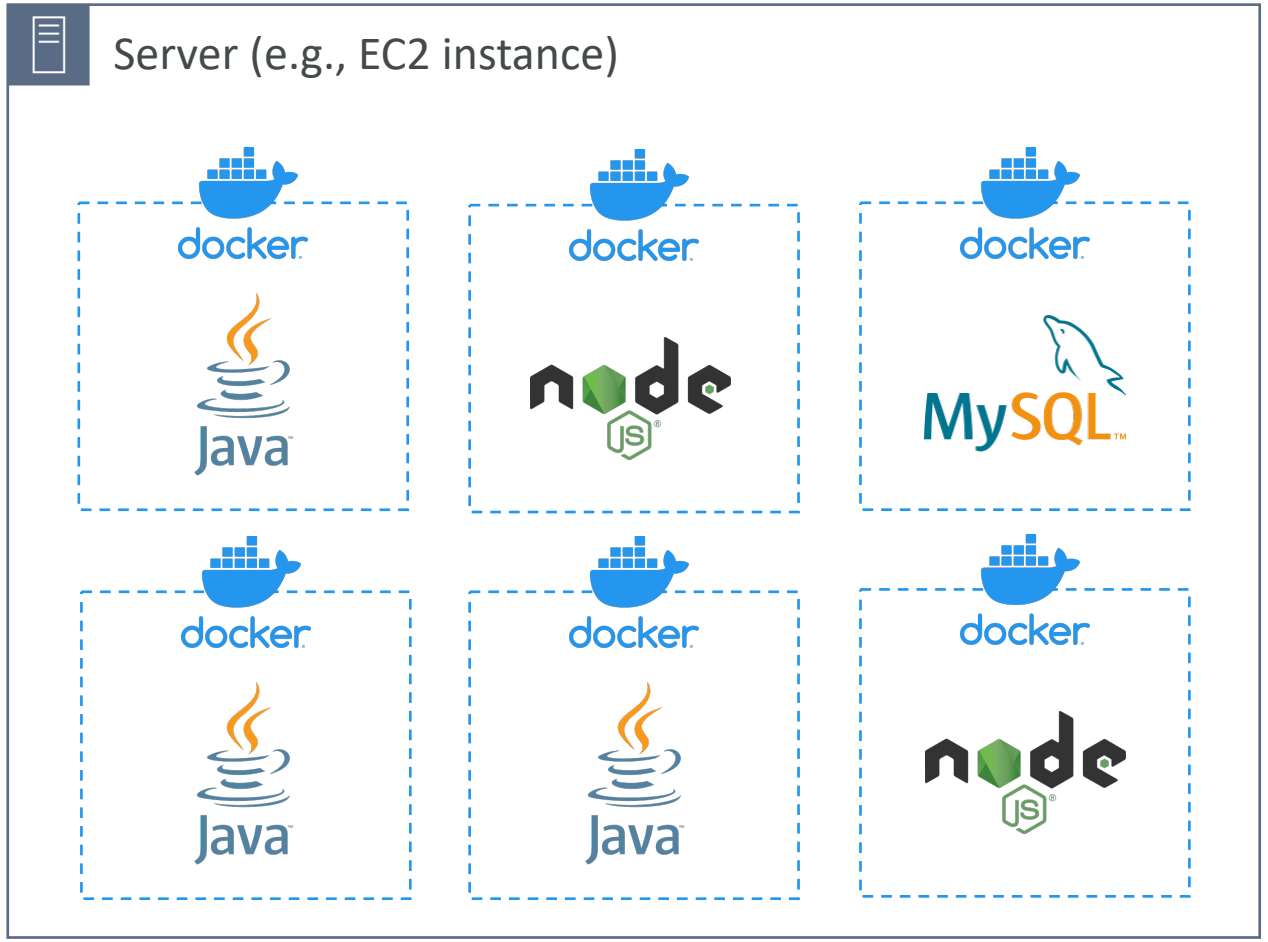


- Docker is a software development platform to deploy apps
- Apps are packaged in **containers** that can be run on any OS
- Apps run the same, regardless of where they're run
  - Any machine
  - No compatibility issues
  - Predictable behavior
  - Less work
  - Easier to maintain and deploy
  - Works with any language, any OS, any technology
- Use cases: microservices architecture, lift-and-shift apps from on-premises to the AWS cloud, …

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Docker on an OS



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Where are Docker images stored?

- Docker images are stored in Docker Repositories

- Docker Hub (https://hub.docker.com)
  - **Public** repository
  - Find base images for many technologies or OS (e.g., Ubuntu, MySQL, …)

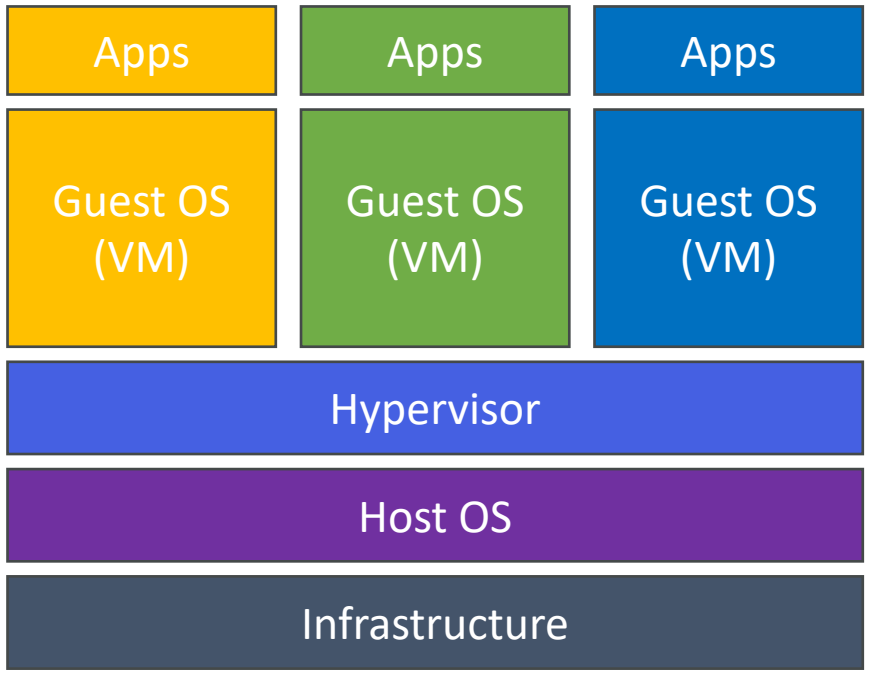
- Amazon ECR (Amazon Elastic Container Registry)
  - **Private** repository
  - **Public** repository (**Amazon ECR Public Gallery** https://gallery.ecr.aws)

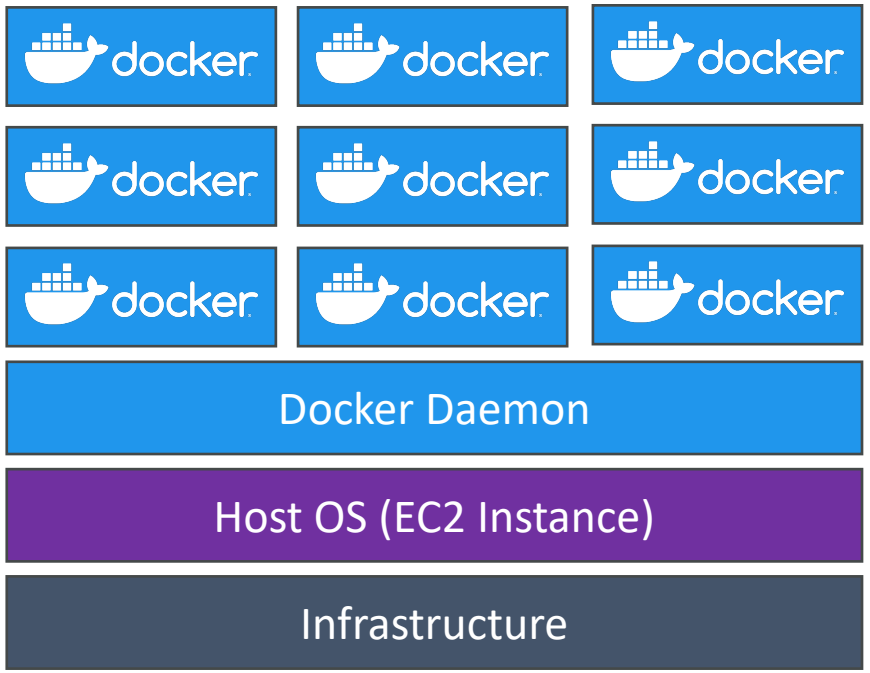
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Docker vs. Virtual Machines

- Docker is "sort of" a virtualization technology, but not exactly
- Resources are shared with the host => many containers on one server

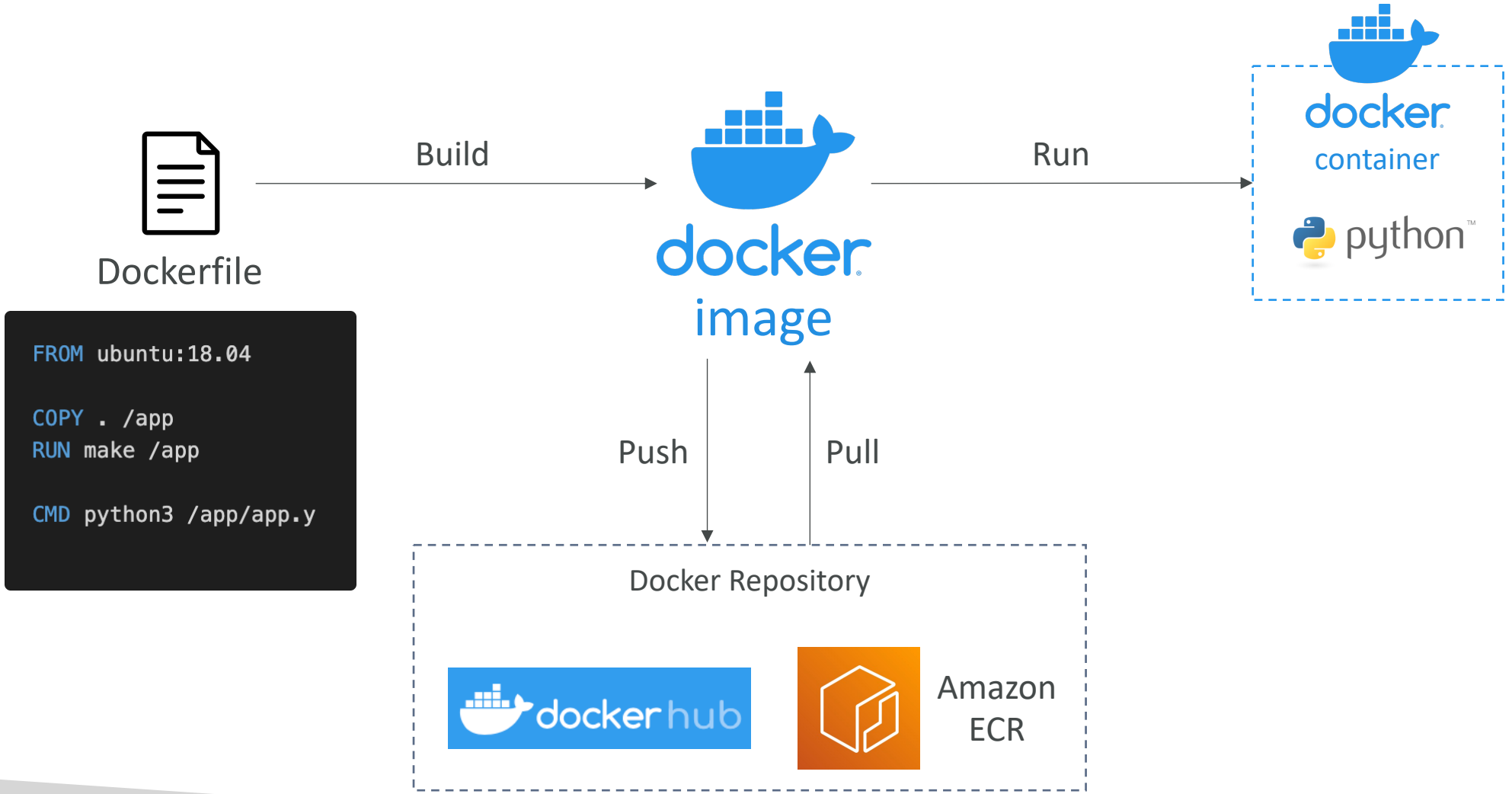




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Getting Started with Docker



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Docker Containers Management on AWS

- Amazon Elastic Container Service (Amazon ECS)
  - Amazon's own container platform



Amazon ECS

- Amazon Elastic Kubernetes Service (Amazon EKS)
  - Amazon's managed Kubernetes (open source)



Amazon EKS

- AWS Fargate
  - Amazon's own Serverless container platform
  - Works with ECS and with EKS



AWS Fargate

- Amazon ECR:
  - Store container images



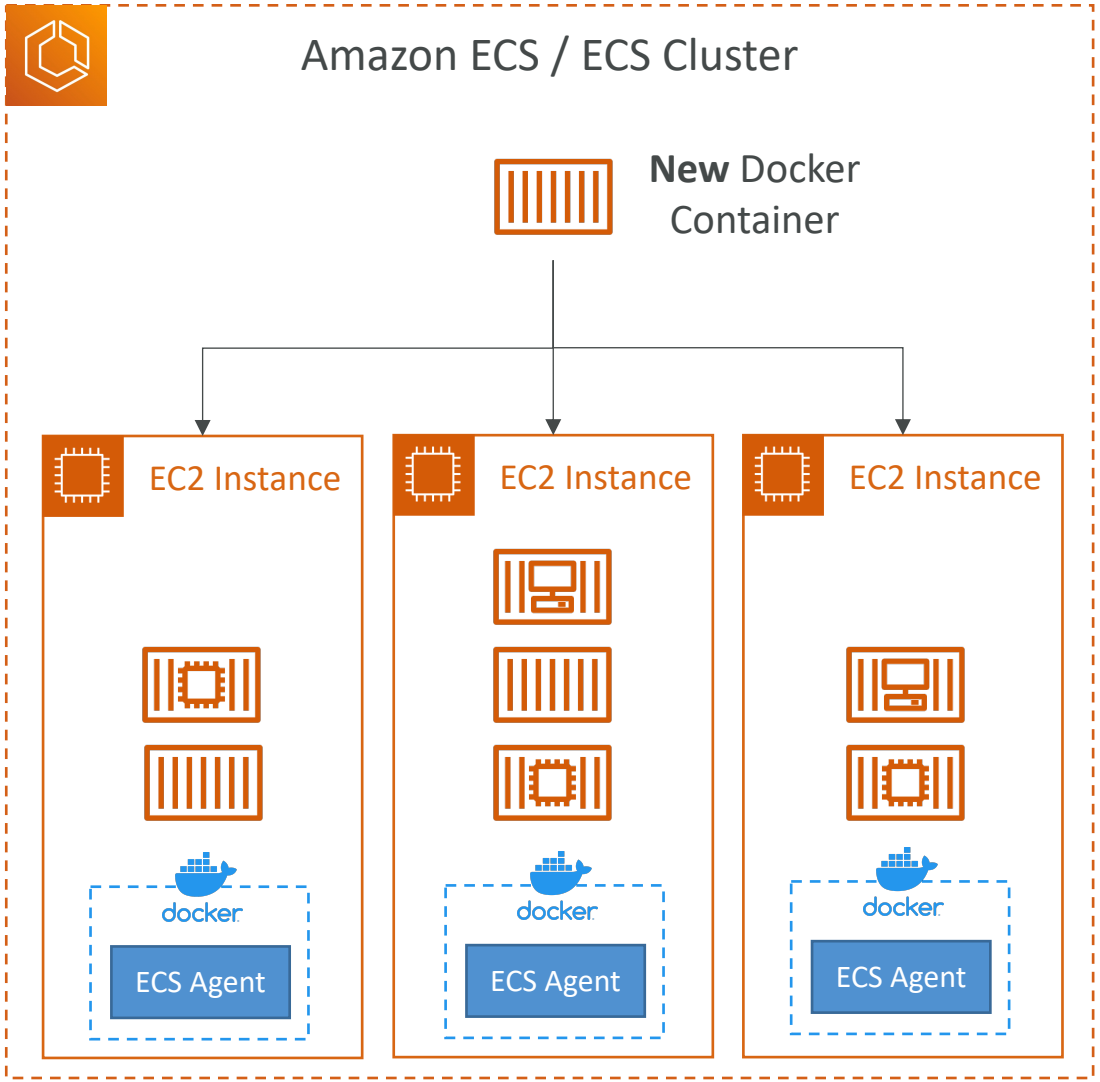
Amazon ECR

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Amazon ECS - EC2 Launch Type

- ECS = Elastic Container Service
- Launch Docker containers on AWS = Launch **ECS Tasks** on ECS Clusters
- **EC2 Launch Type: you must provision & maintain the infrastructure (the EC2 instances)**
- Each EC2 Instance must run the ECS Agent to register in the ECS Cluster
- AWS takes care of starting / stopping containers

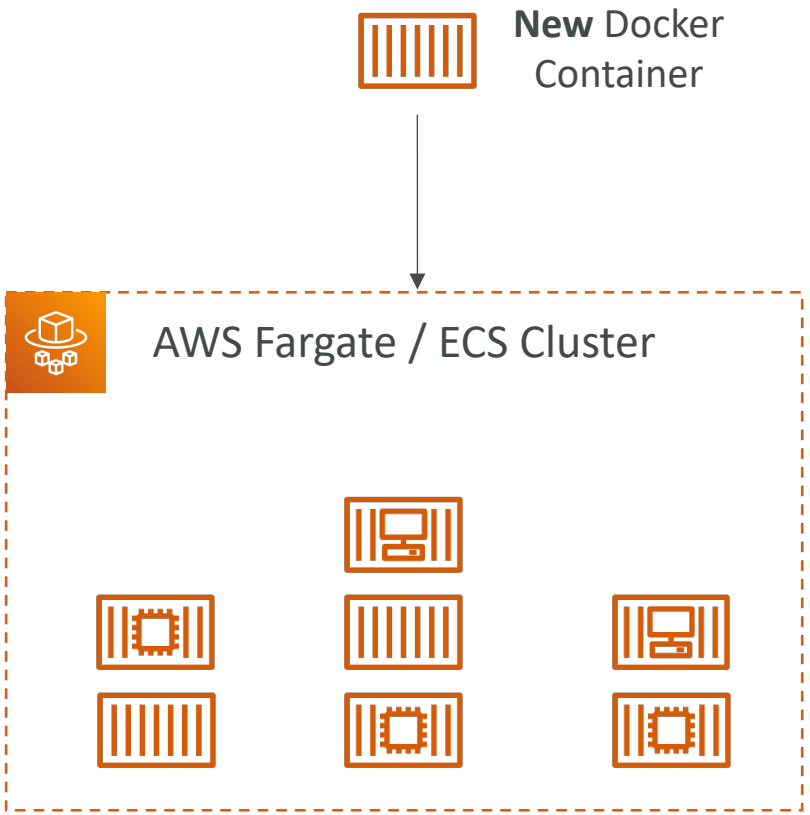


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Amazon ECS – Fargate Launch Type

- Launch Docker containers on AWS
- **You do not provision the infrastructure (no EC2 instances to manage)**
- **It's all Serverless!**
- You just create task definitions
- AWS just runs ECS Tasks for you based on the CPU / RAM you need
- To scale, just increase the number of tasks. Simple - no more EC2 instances

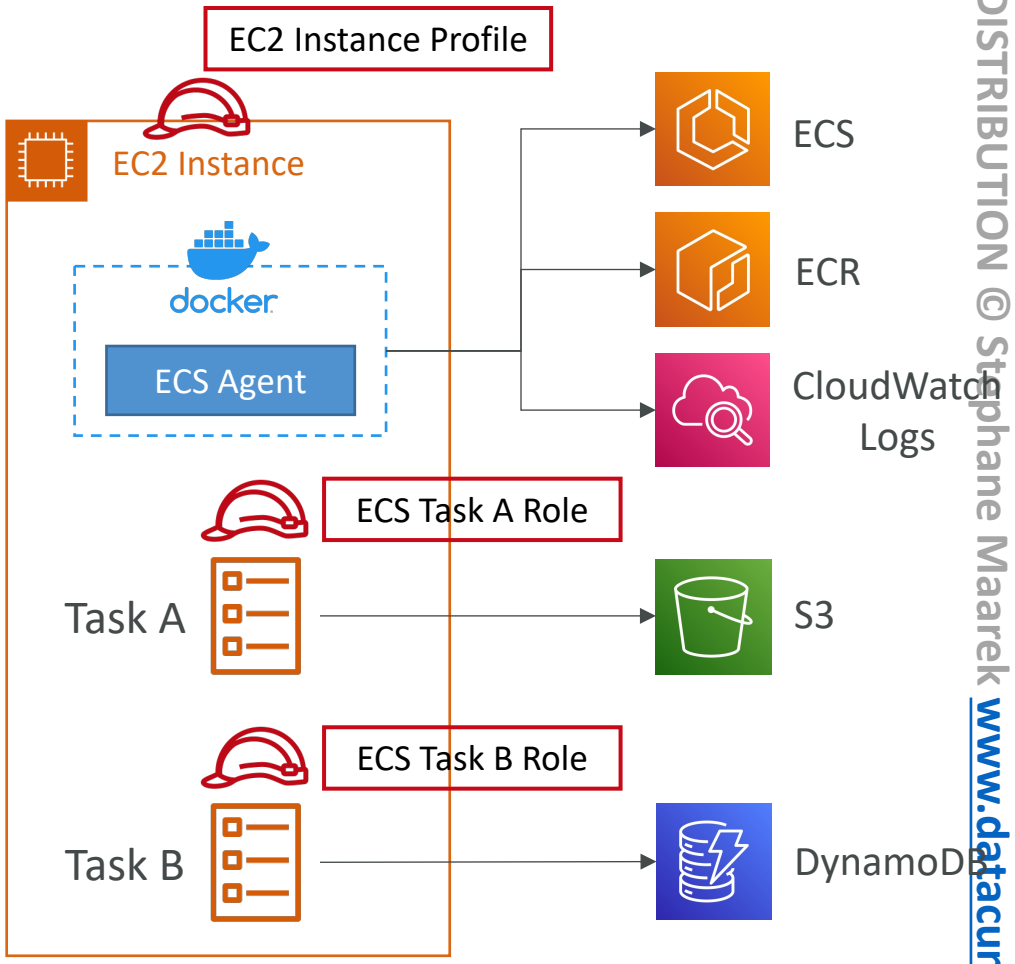


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Amazon ECS – IAM Roles for ECS

- EC2 Instance Profile (EC2 Launch Type only):
  - Used by the ECS agent
  - Makes API calls to ECS service
  - Send container logs to CloudWatch Logs
  - Pull Docker image from ECR
  - Reference sensitive data in Secrets Manager or SSM Parameter Store
- ECS Task Role:
  - Allows each task to have a specific role
  - Use different roles for the different ECS Services you run
  - Task Role is defined in the task definition

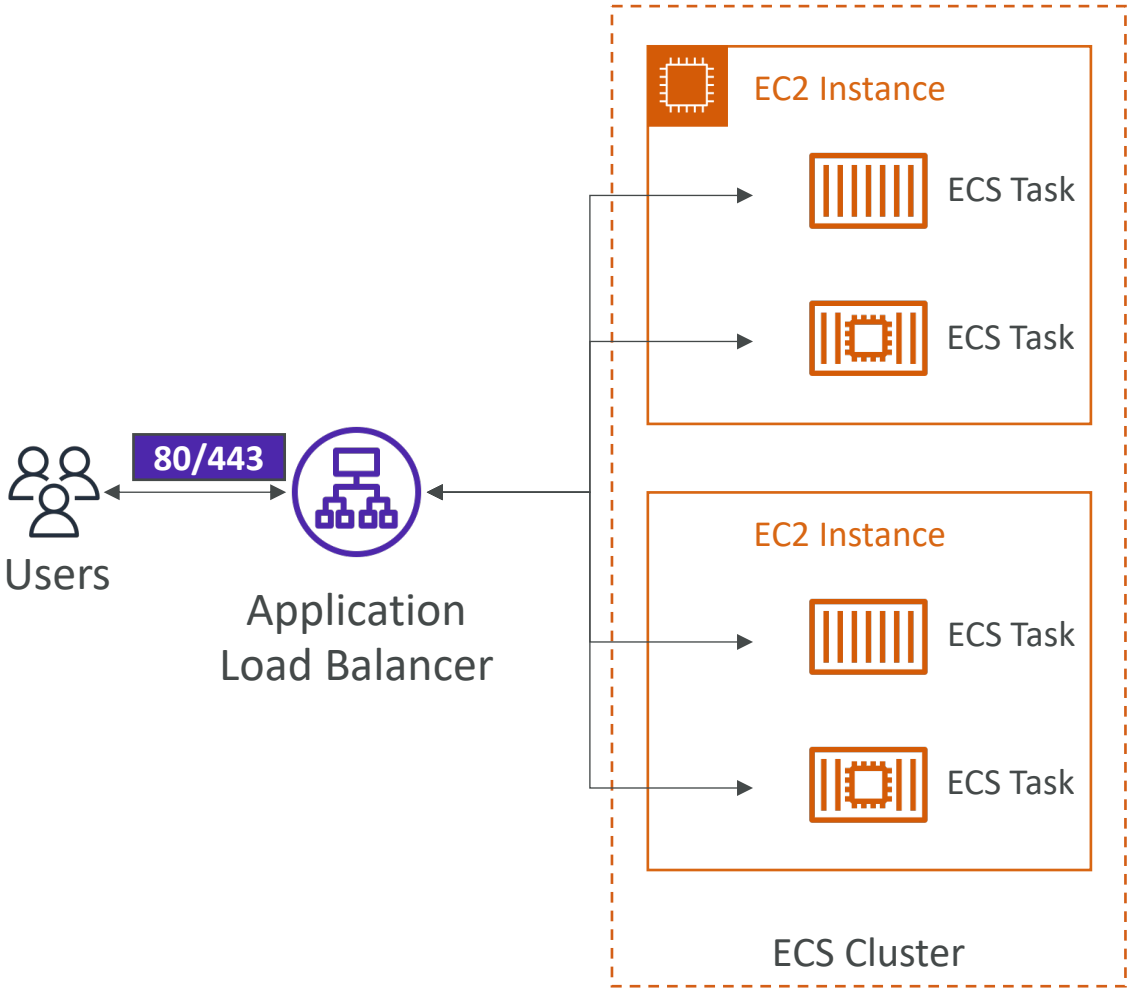


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Amazon ECS – Load Balancer Integrations

- Application Load Balancer supported and works for most use cases
- Network Load Balancer recommended only for high throughput / high performance use cases, or to pair it with AWS Private Link
- Classic Load Balancer supported but not recommended (no advanced features – no Fargate)



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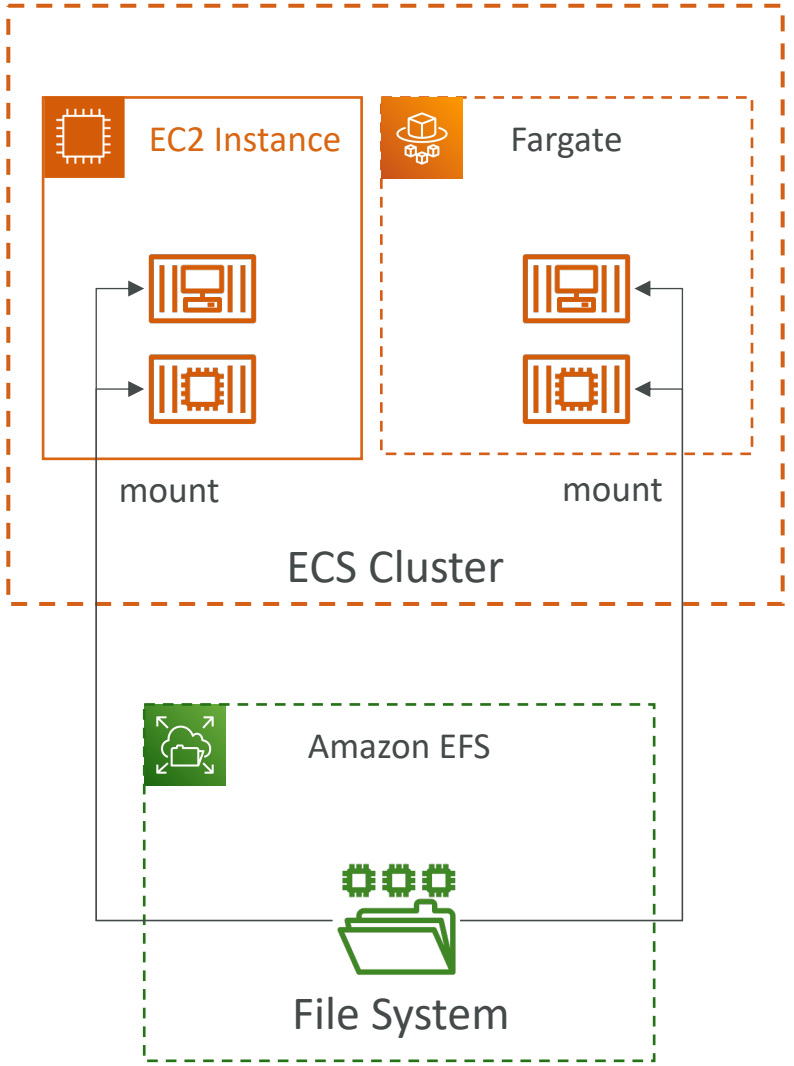
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Amazon ECS – Data Volumes (EFS)

- Mount EFS file systems onto ECS tasks
- Works for both **EC2** and **Fargate** launch types
- Tasks running in any AZ will share the same data in the EFS file system
- **Fargate + EFS = Serverless**

- Use cases: persistent multi-AZ shared storage for your containers

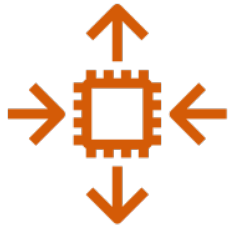
- Note:
  - Amazon S3 cannot be mounted as a file system



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ECS Service Auto Scaling



- Automatically increase/decrease the desired number of ECS tasks

- Amazon ECS Auto Scaling uses **AWS Application Auto Scaling**
  - ECS Service Average CPU Utilization
  - ECS Service Average Memory Utilization - Scale on RAM
  - ALB Request Count Per Target – metric coming from the ALB

- **Target Tracking** – scale based on target value for a specific CloudWatch metric
- **Step Scaling** – scale based on a specified CloudWatch Alarm
- **Scheduled Scaling** – scale based on a specified date/time (predictable changes)

- ECS Service Auto Scaling (task level) ≠ EC2 Auto Scaling (EC2 instance level)
- Fargate Auto Scaling is much easier to setup (because **Serverless**)

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EC2 Launch Type – Auto Scaling EC2 Instances

- Accommodate ECS Service Scaling by adding underlying EC2 Instances

- Auto Scaling Group Scaling
  - Scale your ASG based on CPU Utilization
  - Add EC2 instances over time

- ECS Cluster Capacity Provider
  - Used to automatically provision and scale the infrastructure for your ECS Tasks
  - Capacity Provider paired with an Auto Scaling Group
  - Add EC2 Instances when you're missing capacity (CPU, RAM…)

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